

YARD MATH · PRINTABLE PLANNER

Yard Project Planner

Measure it. Estimate it. Price it. Get it done.

Worksheets for mulch, soil, gravel, concrete, sod and seed, plus quick-reference charts, cost comparisons, supplier quotes and a seasonal checklist.

Project: _____

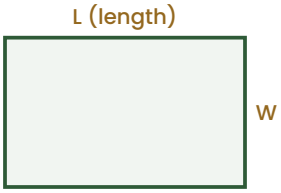
Year: _____

START HERE

How to Measure Irregular Areas

Almost any yard shape can be split into simple pieces. Measure each piece in feet, find its area, then add them up. Round up when you buy, not when you measure.

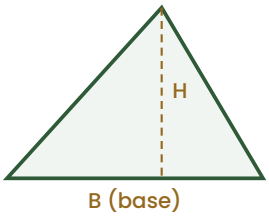
Rectangle



Area = L x W

length times width

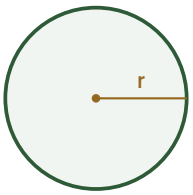
Triangle



Area = 1/2 x B x H

H is square to B

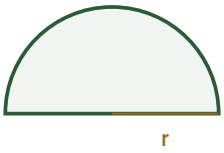
Circle



Area = 3.14 x r x r

r = half the diameter

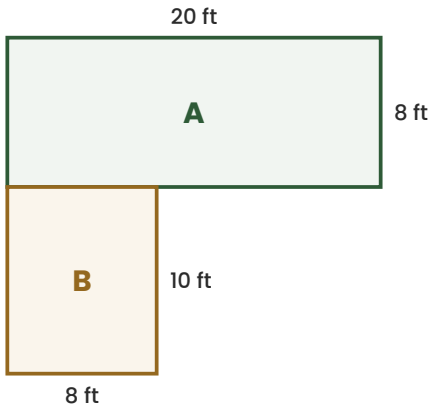
Half circle



Area = 1.57 x r x r

half of a circle

Example: an L-shaped bed



1. **Split** the shape into pieces that do not overlap.
2. **Piece A:** 20 ft x 8 ft = **160 sq ft**
3. **Piece B:** 8 ft x 10 ft = **80 sq ft**
4. **Total:** 160 + 80 = **240 sq ft**
5. **Add extra:** 240 x 1.05 = about **252 sq ft** to buy for (5% waste).

Tips for curvy and odd shapes

- **Offset method for free-form beds:** run a straight tape down the long side. Every 2-5 ft, measure the width across. Average those widths, then multiply by the length.
- **Kidney or oval shapes:** treat as a rectangle plus two half circles, or use the offset method above.
- **Subtract what you skip:** find the outer area, then subtract patios, trees, or a shed that sits inside it.
- **Inches to feet:** divide by 12. 6 in = 0.5 ft, 3 in = 0.25 ft, 9 in = 0.75 ft.
- **Square yards:** sq ft divided by 9. **Acre:** 43,560 sq ft.

PLAN

Project Overview

Write the big picture down before you buy anything. It keeps the budget honest and the weekend realistic.

Project name: _____ **Start date:** _____

Location in yard: _____ **Finish by:** _____

Goals: what should this look like when it's done?

Budget

Item	Estimate $	Actual $	Difference $
Total			

Tip: set aside 10-15% of the budget for delivery fees, extra material, and tool rental.

Timeline

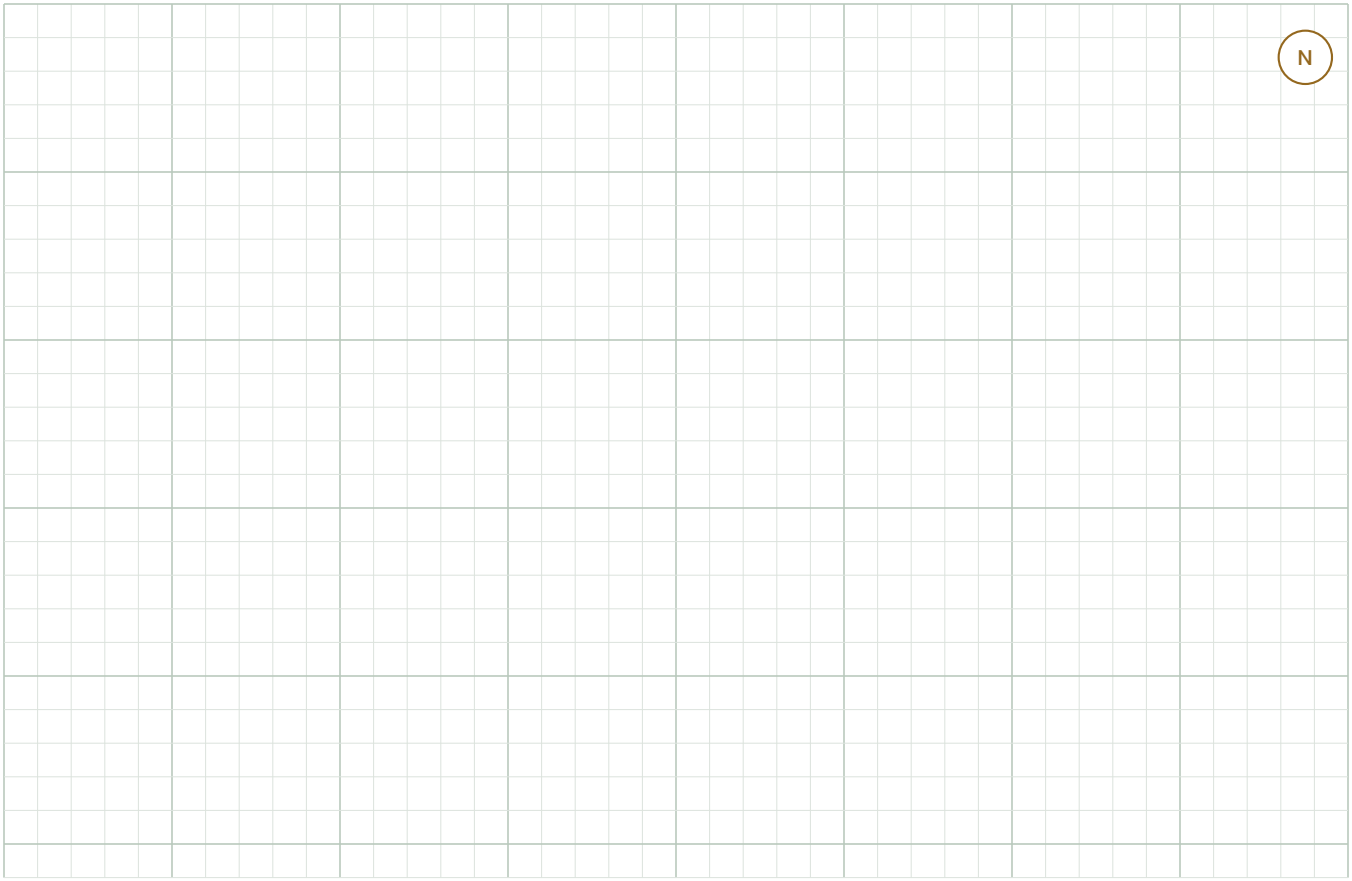
Step	Task	Target date	Done
1			
2			
3			
4			
5			
6			

MEASURE

Measurement Worksheet

Sketch the area on the grid (1 square = 1 ft, or set your own scale). Label each piece, then record it below.

Area / project: _____ **Scale: 1 square =** _____ ft



Piece	Shape	Length (ft)	Width / r (ft)	Area (sq ft)	+ or − ?
A					
B					
C					
D					
E					
Total					

ESTIMATE

Mulch, Soil & Compost

Use this sheet for mulch, topsoil, garden soil, compost, or raised-bed fill.

Project / area: _____ **Date:** _____

FORMULAS

Cubic yards = area (sq ft) x depth (in) ÷ 324
Cubic feet = cubic yards x 27
Bags = cubic feet ÷ bag size (cu ft). Round up.
Example: 240 sq ft x 3 in ÷ 324 = 2.22 cu yd = 60 cu ft = 30 bags of 2 cu ft.
Typical depths: mulch 2-3 in; topdressing compost 0.25-0.5 in; new bed soil 4-8 in; raised beds 8-12 in.

1. Area

Piece	Length (ft)	Width (ft)	Area (sq ft)	Depth (in)
A				
B				
C				
Total				

2. Calculate

Material: _____

Total area: _____ sq ft

Depth: _____ in

Cubic yards: _____ cu yd

+ extra (5-10%): _____ cu yd

Cubic feet (x 27): _____ cu ft

Bag size: _____ cu ft

Bags needed: _____ bags

Price per bag / yd: _____ $

Estimated cost: _____ $

Notes

ESTIMATE

Gravel & Stone

For pea gravel, crushed stone, river rock, decomposed granite, paver base, and sand.

Project / area: _____ **Date:** _____

FORMULAS

Cubic yards = area (sq ft) x depth (in) ÷ 324
Tons = cubic yards x weight (tons per cu yd). Typical gravel is about 1.4-1.7 tons/cu yd.
Bags = cubic yards x 27 ÷ bag size (cu ft). A 0.5 cu ft bag of gravel weighs about 50 lb.
Example: 200 sq ft path x 3 in ÷ 324 = 1.85 cu yd x 1.5 = about 2.8 tons.
Typical depths: decorative stone 2-3 in; walkway 3-4 in; paver base 4-6 in; driveway 8-12 in total.

1. Area

Piece	Length (ft)	Width (ft)	Area (sq ft)	Depth (in)
A				
B				
C				
Total				

2. Calculate

Stone type: _____

Total area: _____ sq ft

Depth: _____ in

Cubic yards: _____ cu yd

+ extra (5-10%): _____ cu yd

Weight per cu yd: _____ tons

Tons needed: _____ tons

Bags (if bagged): _____ bags

Price per ton / bag: _____ $

Estimated cost: _____ $

Most suppliers sell stone by the ton. Ask for their weight per cubic yard; it varies by stone and moisture.

Notes

ESTIMATE

Concrete

For slabs, pads, footings and post holes. Bagged mix is easiest for small jobs; order ready-mix for larger pours.

Project / area: _____ **Date:** _____

FORMULAS

Slab cubic feet = length (ft) x width (ft) x thickness (in) ÷ 12
Cubic yards = cubic feet ÷ 27 (add 5-10% for uneven ground and spills)
Bags = cubic feet ÷ bag yield: 40 lb = 0.30 cu ft; 60 lb = 0.45; 80 lb = 0.60
Round hole cu ft = 3.14 x (diameter in ÷ 24)² x depth in ÷ 12, minus the post
Example: 10 x 10 ft pad, 4 in thick = 33.3 cu ft = 1.23 cu yd = 56 bags of 80 lb.

1. Area

Piece	Length (ft)	Width (ft)	Thick (in)	Cubic feet
A				
B				
Total				

2. Calculate

Total cubic feet: _____ cu ft

+ extra (5-10%): _____ cu ft

Cubic yards: _____ cu yd

Bag size: _____ lb

Bags needed: _____ bags

Price per bag / yd: _____ $

Delivery / pump fee: _____ $

Estimated cost: _____ $

Post holes (optional)

Holes	Hole dia. (in)	Depth (in)	Cu ft each	Total cu ft

Notes

ESTIMATE

Sod & Grass Seed

Plan a new lawn, a patch, or an overseeding job. Fill in the side you need.

Project / area: _____ **Date:** _____

FORMULAS

Lawn area = sum of all pieces (sq ft). Add 5-10% for cuts along curves and edges.
Sod pieces = area ÷ 2.67 for 16 x 24 in slabs (or ÷ your piece size in sq ft)
Sod pallets = area ÷ sq ft per pallet (often 400-500). Round up.
Seed (lb) = area ÷ 1,000 x seeding rate (lb per 1,000 sq ft). See chart on page 10.
Example: 1,800 sq ft tall fescue at 7 lb/1,000 = 12.6 lb; as sod = 4 pallets of 450 sq ft.

1. Area

Piece	Length (ft)	Width (ft)	Area (sq ft)
A			
B			
C			
Total			

2a. Sod

Grass type: _____

Area + extra: _____ sq ft

Piece size: _____ sq ft

Pieces needed: _____ pcs

Sq ft per pallet: _____ sq ft

Pallets needed: _____ pallets

Price per pallet: _____ $

Delivery fee: _____ $

Estimated cost: _____ $

2b. Seed

Grass type: _____

New lawn or overseed? _____

Area: _____ sq ft

Rate per 1,000 sq ft: _____ lb

Seed needed: _____ lb

Bag size: _____ lb

Bags needed: _____ bags

Price per bag: _____ $

Estimated cost: _____ $

Notes

QUICK REFERENCE

Charts: Mulch, Bags & Weights

Rounded figures for planning. Product labels and your supplier's numbers always win.

Coverage of 1 cubic yard

Depth	Covers about
1 in	324 sq ft
2 in	162 sq ft
3 in	108 sq ft
4 in	81 sq ft
6 in	54 sq ft
12 in	27 sq ft

Works for any loose material: mulch, soil, compost, gravel. Coverage = 324 ÷ depth in inches.

Bags per cubic yard

Bag size	Bags per cu yd
0.5 cu ft	54
0.75 cu ft	36
1 cu ft	27
1.5 cu ft	18
2 cu ft	13.5 (buy 14)
3 cu ft	9

1 cubic yard = 27 cubic feet. Bags per yard = 27 ÷ bag size.

Approximate weight per cubic yard

Material	Pounds per cu yd	Tons	Notes
Wood mulch / bark	400-800	0.2-0.4	Heavier when wet
Compost	800-1,400	0.4-0.7	Varies with moisture
Topsoil (loose, dry)	2,000-2,600	1.0-1.3	Wet soil: up to ~1.7 tons
Sand (dry)	2,600-3,000	1.3-1.5	Wet sand: ~1.5-1.7 tons
Gravel / crushed stone	2,800-3,400	1.4-1.7	Pea gravel, #57, base
River rock (2-6 in)	2,700-3,400	1.4-1.7	Larger rock, more air gaps
Decomposed granite	2,700-3,000	1.35-1.5	Before compaction
Concrete (mixed, wet)	3,700-4,200	1.85-2.1	About 4,000 lb typical

Truck check: a typical half-ton pickup can safely carry about 1 cu yd of mulch or compost, but often only about 1/2 cu yd of soil or gravel. Check your vehicle's payload rating before loading.

QUICK REFERENCE

Charts: Concrete, Sod & Seed

Rounded figures for planning. Product labels and your supplier's numbers always win.

Bagged concrete yield (standard concrete mix)

Bag	Yield (cu ft)	Bags per cu yd	4 in slab covers	Holes 8 in x 2 ft*
40 lb	0.30	90	0.9 sq ft	about 2.3 bags
60 lb	0.45	60	1.35 sq ft	about 1.6 bags
80 lb	0.60	45	1.8 sq ft	about 1.2 bags

*Per hole, 8 in diameter x 24 in deep (0.70 cu ft), before subtracting the post. Yields per Quikrete data sheet; other brands are similar.

Sod: common sizes

Format	Piece size	Sq ft each	Per pallet
Slab (common in South)	16 x 24 in	2.67	about 150-190 slabs
Small roll (common in North)	about 18 in x 40 in	about 5	about 80-100 rolls
Pallet coverage	-	-	400, 450 or 500 sq ft
Big roll (machine laid)	about 42 in x 100 ft	varies	by order

Piece and pallet sizes vary by farm. Always ask the sq ft per pallet. Lay sod within 24 hours of delivery.

Grass seed: new-lawn rates

Species	Season	New lawn (lb/1,000 sq ft)	Overseed (lb/1,000 sq ft)
Kentucky bluegrass	Cool	1.5-2	about 1
Tall fescue	Cool	5-8	3-4
Perennial ryegrass	Cool	4-6	2-3
Fine fescue	Cool	3-5	2-3
Bermudagrass (hulled)	Warm	0.5-1	-
Zoysiagrass	Warm	1-2	-
Centipedegrass	Warm	0.25-0.5	-

New-lawn ranges follow US university extension guides (e.g. Univ. of Tennessee, Purdue, Alabama Extension). Overseed rates are a common rule of thumb (about half). Follow the rate on your seed bag; blends and coated seed differ. Warm-season lawns are usually not overseeded with the same species.

COMPARE

Bag vs. Bulk Cost Comparison

Bulk is usually cheaper per yard for big jobs, but delivery fees can flip the math on small ones. Price both.

Material: _____ **Amount needed:** _____ cu yd

Bagged

Store: _____

Bag size: _____ cu ft

Bags per cu yd (27 ÷ size): _____

Bags needed: _____ bags

Price per bag: _____ $

Subtotal: _____ $

Sales tax: _____ $

Delivery / trips: _____ $

Total bagged: _____ $

Cost per cu yd: _____ $

Bulk

Supplier: _____

Price per cu yd / ton: _____ $

Minimum order: _____ cu yd

Yards / tons ordered: _____

Material subtotal: _____ $

Sales tax: _____ $

Delivery fee: _____ $

Tarp / extras: _____ $

Total bulk: _____ $

Cost per cu yd: _____ $

Decision

☐ Bagged wins ☐ Bulk wins **Savings:** _____ $

RULES OF THUMB

Under about 1-2 cu yd, bags are often cheaper once delivery is counted. Over 3 cu yd, bulk usually wins.
Bags are easier to carry to a back yard. Bulk needs a spot for the pile and a tarp if it sits.
Ask about bag sales (e.g. 5 for $10) and contractor pricing on pallets of bags.

COMPARE

Supplier Quote Comparison

Call or visit at least three suppliers. Ask the same questions so the quotes are apples to apples.

Material & amount: _____ **Needed by:** _____

	Supplier 1	Supplier 2	Supplier 3
Contact / phone			
Product name			
Price per yd / ton / bag			
Quantity			
Material subtotal			
Delivery fee			
Minimum order			
Tax			
Total quoted			
Delivery date			
Drop location OK?			
Quote good until			
Notes			

Questions to ask

- ☐ Is the price per cubic yard or per ton?
- ☐ Can they dump it on the driveway, or only at the curb?
- ☐ Is there a short-load fee for small orders?
- ☐ Is delivery included? Is there a fuel charge?
- ☐ Do they screen or test the soil / compost?
- ☐ Can I see or pick up a sample first?

MAINTAIN

Seasonal Yard Checklist

Timing depends on your climate. Cool-season lawns (fescue, bluegrass) peak in spring and fall; warm-season lawns (bermuda, zoysia) in summer.

Spring

- [ ] Clean up beds, edges and winter debris
- [ ] Test soil (every 2-3 years)
- [ ] Sharpen mower blade; tune up equipment
- [ ] Apply pre-emergent before soil hits 55°F
- [ ] Refresh mulch to 2-3 in total depth
- [ ] Top-dress beds with compost
- [ ] Check irrigation for leaks
- [ ] Plant trees, shrubs and perennials
- [ ] _____
- [ ] _____

Summer

- [ ] Mow high; never cut more than 1/3 of the blade
- [ ] Water deeply, early morning, 1 in per week
- [ ] Seed or sod warm-season grass
- [ ] Pull weeds before they set seed
- [ ] Watch for grubs and brown patches
- [ ] Deadhead flowers; stake tall plants
- [ ] Build patios, paths and walls (dry weather)
- [ ] Keep mulch away from stems and trunks
- [ ] _____
- [ ] _____

Fall

- [ ] Seed or overseed cool-season grass
- [ ] Core aerate compacted lawns
- [ ] Fertilize cool-season lawns
- [ ] Mulch or remove fallen leaves
- [ ] Plant spring bulbs, trees and shrubs
- [ ] Add compost to empty beds
- [ ] Drain hoses; blow out irrigation
- [ ] Final low mow before dormancy
- [ ] _____
- [ ] _____

Winter

- [ ] Clean, sharpen and store tools
- [ ] Winterize the mower (fuel, oil)
- [ ] Prune dormant trees and shrubs
- [ ] Stay off frozen or soggy turf
- [ ] Use plant-safe de-icer near beds
- [ ] Sketch next year's projects
- [ ] Price materials; watch for off-season deals
- [ ] Order seeds and plan the garden
- [ ] _____
- [ ] _____

Notes & Sketches